Parking Space Availability Prediction Using Random Forest and Time-Series Models

projectreportsubmittedinpartialfulfilmentoftherequirementsfortheCourse

MACHINELEARNING

BACHELOROFTECHNOLOGY

in

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by

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DEPARTMENT OF CSE

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1. Abstract

The increasing number of vehicles in urban areas has made parking management a significa challenge, resulting in traffic congestion, fuel wastage, environmental pollution, and inconvenience t drivers searching for available parking spaces. This project proposes a machine learning-based approach fo predicting parking space availability by combining Random Forest and Time-Series models to provic accurate and reliable parking occupancy predictions. The Random Forest model is used to learn th relationship between parking occupancy and influencing factors such as date, time, day of the week, weathe conditions, holidays, and previous occupancy records, while the Time-Series model captures tempor patterns to forecast future parking availability. The proposed system is trained and evaluated using th Parking Occupancy Detection dataset obtained from Kaggle, which contains historical parking occupanc information collected over different time intervals. Data preprocessing techniques, including missing valu handling, feature selection, categorical encoding, and normalization, are applied before training th prediction models to improve performance and reliability. The developed system aims to provide users wi real-time parking availability predictions, thereby reducing parking search time and improving th utilization of available parking spaces. The performance of the proposed model is evaluated using accurac precision, recall, F1-score, ROC-AUC, confusion matrix, mean absolute error (MAE), and root mean squa error (RMSE), providing a comprehensive assessment of both classification and forecasting performanc under different parking scenarios.

2. INTRODUCTION:

The rapid growth of urbanization and the continuous increase in the number of vehicles have create serious parking challenges in modern cities. Drivers often spend a considerable amount of time searchin for vacant parking spaces, which contributes to increased traffic congestion, unnecessary fuel consumptio environmental pollution, and delays in reaching their destinations. Existing parking systems generall provide only the current occupancy status and do not assist users in predicting future parking availabilit Motivated by these challenges and the growing need for intelligent transportation systems, this proje focuses on developing a smart parking prediction system capable of providing accurate parking availabilit information before a driver reaches the parking location.

Machine learning has become an effective solution for solving prediction problems by learnin patterns from historical data. Random Forest is widely used because of its high prediction accurac robustness, and ability to handle complex relationships among multiple features, while Time-Series analys is effective in identifying temporal trends and forecasting future occupancy based on historical observation This project integrates both techniques into a unified prediction framework that analyzes historical parkin occupancy data together with features such as date, time, day of the week, weather conditions, holidays, an previous occupancy records. The complete machine learning lifecycle, including data collectio preprocessing, feature engineering, model training, evaluation, deployment, and monitoring, is followed t develop a practical parking prediction system capable of supporting smart parking management an improving urban transportation efficiency.

3. SOFTWAREREQUIREMENTS:

The proposed parking space prediction system is implemented using a Python-based machine learnin environment supported by data processing, visualization, and web application development tools. Th development environment provides facilities for preprocessing the parking occupancy dataset, training an evaluating prediction models, visualizing the obtained results, and deploying the trained model as a simp. web application for real-time prediction. A standard computer with sufficient processing capability an memory is adequate for developing, testing, and deploying the proposed system. The software and hardwai requirements used for the project are summarized below.

Category	Requirement
Operating System	Windows 10/11 or ubuntu 20.04 and above
Programming language	Python 3.9 or above
Machine learning library	Scikit-learn
Time-Series Library	Statsmodels
Supporting Libraries	NumPy,pandas,Matploilb,seaborn
Development Enviroment	Jupyter Notebook/VS Code
Processor	Intel core i5 or equivalent
RAM	8 GB minimum
Storage	20 GB free disk space
Web Browser	Google chrome/microsoft edge

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